

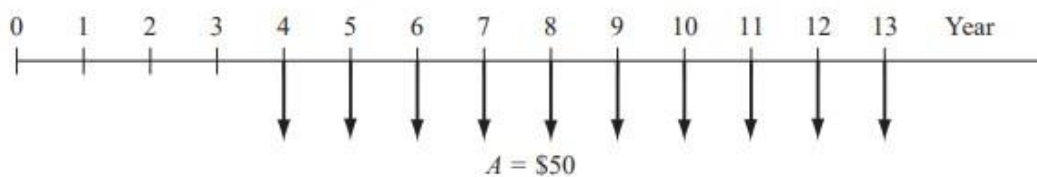
Chapter 3 Combining Factors and Spreadsheet Functions

• Calculations for Uniform Series That Are Shifted

When a uniform series begins at a time other than at the end of period 1, it is called a shifted series. In this case several methods can be used to find the equivalent present worth P . For example, P of the uniform series shown in the figure below could be determined by any of the following methods:

- Use the P/F factor to find the present worth of each disbursement at year 0 and add them.
- Use the F/P factor to find the future worth of each disbursement in year 13, add them, and then find the present worth of the total, using $P = F (P/F, i, 13)$.
- Use the F/A factor to find the future amount $F = A (F/A, i, 10)$, and then compute the present worth, using $P = F (P/F, i, 13)$.
- Use the P/A factor to compute the “present worth” $P_3 = A (P/A, i, 10)$ (which will be located in year 3, not year 0), and then find the present worth in year 0 by using the $(P/F, i, 3)$ factor.

Typically, the last method is used for calculating the present worth of a uniform series that does not begin at the end of period 1.



It is extremely important to remember that:

- The present worth is always located one period prior to the first uniform series amount when using the P/A factor.
- The future worth is always located in the same period as the last

uniform series amount when using the F/A factor.

Spreadsheet Functions

The more complex that cash flow series become, the more useful are the spreadsheet functions. When the uniform series A is shifted, the NPV function is used to determine P, and the PMT function finds the equivalent A value. The NPV function, like the PV function, determines the P values, but NPV can handle any combination of cash flows directly from the cells. Enter the net cash flows in contiguous cells (column or row), making sure to enter “0” for all zero cash flows. Use the format

NPV (i%, second_cell: last_cell) + first_cell

First_cell contains the cash flow for year 0 and must be listed separately for NPV to correctly account for the time value of money. The cash flow in year 0 may be 0. The easiest way to find an equivalent A over n years for a shifted series is with the PMT function, where the P value is from the NPV function above. The format is the same as we learned earlier; the entry for P is a cell reference, not a number.

PMT (i %, n, cell_with_P, F)

Alternatively, the same technique can be used when an F value was obtained using the FV function. Now the last entry in PMT is “cell_with_F.”

It is very fortunate that any parameter in a spreadsheet function can itself be a function. Thus, it is possible to write the PMT function in a single cell by embedding the NPV function (and FV function, if needed). The format is

PMT (i %, n, NPV (i %, second_cell: last_cell) + first_cell, F)

Of course, the answer for A is the same for the two-cell operation or a single-cell, embedded function.

- **Calculations Involving Uniform Series and Randomly Placed Single Amounts**

When a cash flow includes both a uniform series and randomly placed single amounts, the procedures of the previous section are applied to the uniform series and the single-amount formulas are applied to the one-time cash flows. This approach, is merely a combination of previous ones.

When you calculate the A value for a cash flow series that includes randomly placed single amounts and uniform series, first convert everything to a present worth or a future worth. Then you obtain the A value by multiplying P or F by the appropriate A/P or A/F factor.

- **Calculations for Shifted Gradients**

In Chapter 2, we derived the relation $P = G (P/G, i, n)$ to determine the present worth of the arithmetic gradient series. The P/G factor, was derived for a present worth in year 0 with the gradient first appearing in year 2.

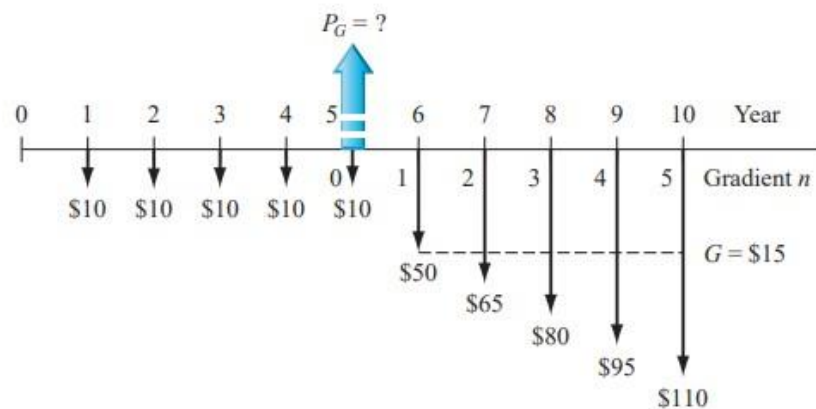
Note: The present worth of an arithmetic gradient will always be located two periods before the gradient starts.

The relation $A = G (A/G, i, n)$ was also derived in chapter 2. The A/G factor in the equation performs the equivalence transformation of a gradient only into an A series from years 1 through n. Recall that the base amount must be treated separately. Then the equivalent P or A values can be summed to obtain the equivalent total present worth P_T and total annual series A_T .

A conventional gradient series starts between periods 1 and 2 of the cash flow sequence. A gradient starting at any other time is called a shifted gradient. The n value in the P/G and A/G factors for a shifted gradient is determined by renumbering the time scale. The period in which the gradient first appears is labeled period 2. The n value for the gradient factor is determined by the renumbered period where the last gradient increase occurs.

Partitioning a cash flow series into the arithmetic gradient series and the remainder of the cash flows can make very clear what the gradient n value should be.

It is important to note that the A/G factor cannot be used to find an equivalent A value in periods 1 through n for cash flows involving a shifted gradient. Consider the cash flow diagram of the figure below. To find the equivalent annual series in years 1 through 10 for the gradient series only, first find the present worth P_G of the gradient in actual year 5, take this present worth back to year 0, and annualize the present worth for 10 years with the A/P factor. If you apply the annual series gradient factor ($A/G, i, 5$) directly, the gradient is converted into an equivalent annual series over years 6 through 10 only, not years 1 through 10, as requested.



To find the equivalent A series of a shifted gradient through all the n periods, first find the present worth of the gradient at actual time 0, then apply the $(A/P, i, n)$ factor.

In chapter 2, we also derived the relation $P_g = A_1 (P/A, g, i, n)$ to determine the present worth of a geometric gradient series, including the initial amount A_1 . The factor was derived to find the present worth in year 0, with A_1 in year 1 and the first gradient appearing in year 2.

The present worth of a geometric gradient series will always be located two periods before the gradient stars, and the initial amount is included in the resulting present worth.

Decreasing arithmetic and geometric gradients are common, and they are often shifted gradient series. That is, the constant gradient is $-G$ or the percentage change is $-g$ from one period to the next, and the first appearance of the gradient is at some time period (year) other than year 2 of the series.

Equivalence computations for present worth P and annual worth A are basically the same as discussed in Chapter 2, except for the following.

For shifted, decreasing gradients:

- The base amount A (arithmetic) or initial amount A_1 (geometric) is the largest amount in the first year of the series.
- The gradient amount is subtracted from the previous year's amount, not added to it.
- The amount used in the factors is $-G$ for arithmetic and $-g$ for geometric gradient series.
- The present worth P_G or P_g is located 2 years prior to the appearance of the first gradient; however, a P/F factor is necessary to find the present worth in year 0